embo

SYSTEM MANAGEMENT

The complete guide: how Embo works, and why.

Every screen, every button, every rule the scheduler follows — with real examples from the sample December 2026 roster, and step-by-step recipes for changing and manipulating everything.



covered, and rested

INSIDE

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HOW TO READ THIS GUIDE

Chapters 1–4 give you the mental model. Chapters 5–7 explain **how the scheduler thinks**. Chapters 8–14 walk through **every dashboard and control**. Chapters 15–16 are practical **how-to recipes**. Coloured boxes mark the type of information:

WHY

The reasoning behind a design choice.

EXAMPLE

A concrete case from the sample data.

TRY IT

Clicks you can do yourself right now.

WATCH OUT

Common mistakes and surprises.

All numbers and screenshots come from the sample dataset with the roster for **December 2026** generated fresh. If you have edited data in your browser, your numbers will differ.

CHAPTER 1

What Embo is, in one page

Embo is a browser-based roster planner for an operations team that supports many D2C brands. It answers one question every month: **“Who works which day, so that every brand gets the coverage it is promised — while everyone still gets their days off?”**

Doing this by hand gets hard fast. With 32 people, 25 brands and 57 brand-processes, a single weekly off can leave a brand with nobody answering its customers, and one sick day can affect several brands at once. Embo does three jobs:

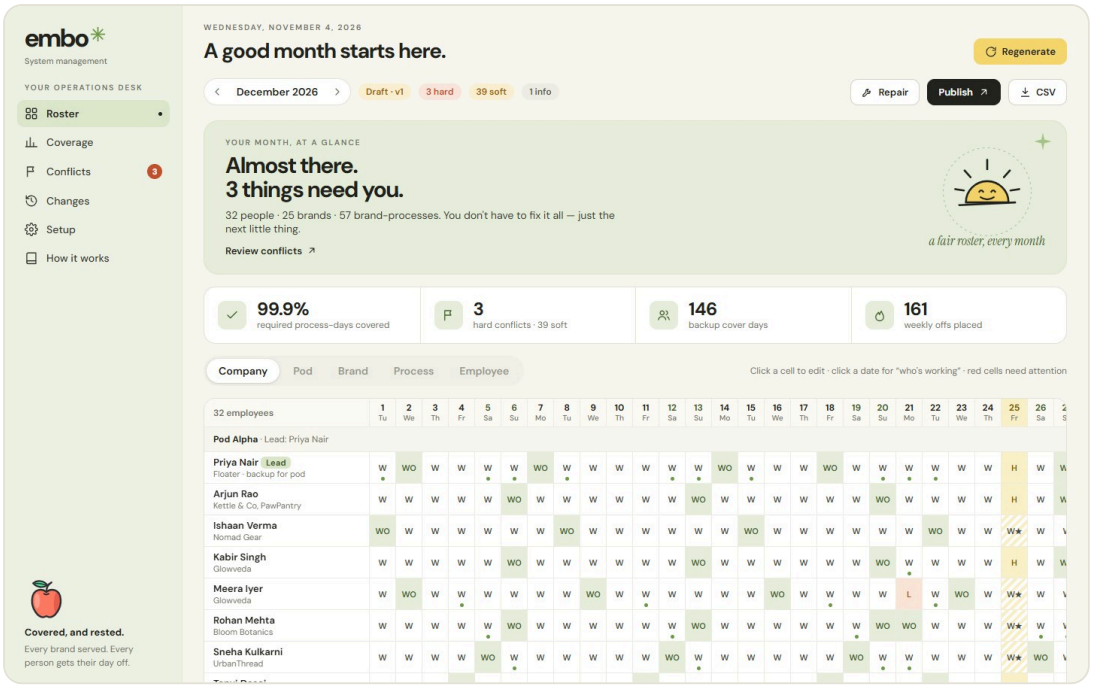
1 · Plan
Generates the month from rules: brand calendars, minimum staffing, holidays, leave, preferences.

→

2 · Check
An independent validator re-checks every rule after every action and lists problems in plain English.

→

3 · Adapt
When life happens (sickness, new rules), it simulates the impact and repairs only what broke.



The Roster dashboard right after generating December 2026. The banner says “3 things need you”: Embo couldn’t satisfy every rule and tells you so instead of hiding it.

WHY IT IS BUILT THIS WAY

- **Solve the scheduling problem, not just draw a calendar.** Every cell is produced by rules and every rule is checked.
- **Never silently produce an invalid roster.** If coverage can’t be met, Embo leaves the problem visible and explains it.
- **Humans keep the judgement calls.** Embo suggests and simulates; the operations manager approves leave, applies fixes and publishes.

CHAPTER 2

Key ideas & vocabulary

The brand-process: the unit of coverage

Embo doesn't staff "brands" — it staffs **brand-processes**. A brand-process is one process run for one brand, for example **Lumen Skincare · CS** (Customer Support) or **Lumen Skincare · ACC** (Abandoned Cart Conversion). Each brand-process carries its own contract:

SETTING	WHAT IT MEANS	EXAMPLE
Operating calendar	Which days the process needs people.	Lumen CS: 365 days. Lumen ACC: Mon–Sat, closed on holidays.
Minimum staff	People needed on an operating day.	PureSip CS: 2 people.
Per-weekday minimum	Overrides the minimum for specific weekdays.	PureSip CS: only 1 on Sundays. Glowveda CS: 2 on Mondays.
Shift	GENERAL (day) or LATE. One person can't do both on the same day.	Nomad Gear CS and Trailblaze are LATE.

WHY BRAND-PROCESS AND NOT BRAND?

Because the brief's hardest cases live inside one brand: "Customer Support runs seven days a week while the brand's other process follows a different schedule." Modelling each process separately lets one brand have different calendars, minimums and people per process.

The four operating calendars

CALENDAR	OPERATES ON	EXAMPLE: FRI 25 DEC (CHRISTMAS) AND SUN 27 DEC
365 days	Every day, including Sundays and public holidays.	Open on both.
Every day except public holidays	All days except holidays.	Closed 25 Dec, open 27 Dec.
Mon–Sat, closed on holidays	Monday to Saturday, not on holidays.	Closed on both.
Custom	Pick weekdays; choose whether holidays close it.	FitFuel Retention: Mon–Fri, closed on holidays.

Owners and backups

People are linked to brand-processes by **assignments** of two types:

- **Owner (PRIMARY)** — does the work. When an owner is working, they count towards that process's coverage automatically.

- **Backup** — trained to step in. A backup only counts when the roster explicitly gives them a **cover** for that day (shown as a green dot in the grid).

Embo also recognises **cross-trained** colleagues: someone who works the same brand on another process, or the same process for another brand in the same pod. They are never used automatically, only *suggested*, and using them raises a soft warning.

EXAMPLE

Bloom Botanics has a single owner, Rohan Mehta, for CS, ACC and NDR. His backups are Sneha Kulkarni (his buddy, who owns UrbanThread) and Priya Nair (Pod Alpha lead). When Rohan takes Sunday 13 Dec off, Sneha covers Bloom CS and NDR; ACC is closed on Sundays so needs nobody.

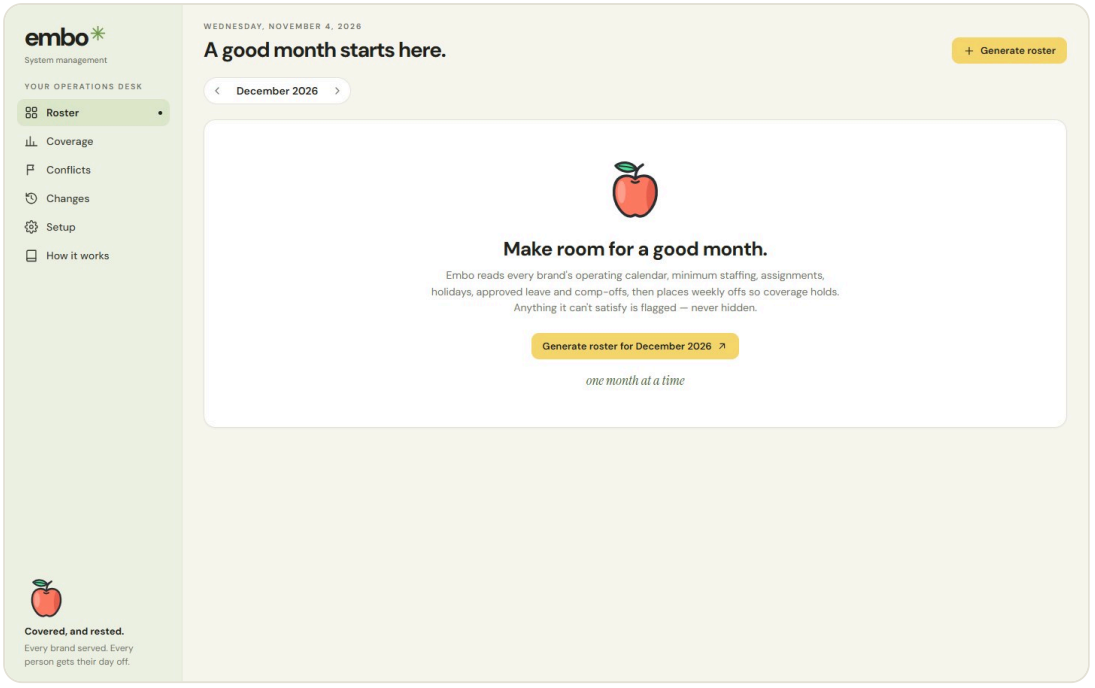
Day statuses (the codes in the grid)

CODE	STATUS	MEANING
W	Working	Works their own brand-processes. A green dot below means they also cover for someone.
W★	Works a holiday	Working on a public holiday — earns one comp-off.
WO	Weekly off	Their rest day for that week.
–	Brands closed	Off because none of their own brand-processes operate that day *and* it wasn't needed as the weekly off — e.g. a Mon–Fri-only owner on Saturday when Sunday is already their weekly off.
L	Leave	Approved planned or sick leave.
CO	Comp-off	Using a compensatory day earned by working a holiday.
H	Holiday	Public holiday off.

Cell markings: a **red outline** = hard issue on that cell; an **orange outline** = soft issue; a **grey corner triangle** = locked manual edit; an **orange dot** = changed since the roster was published.

CHAPTER 3

Getting around the screen



Before a roster exists. Every month has its own roster; a new month starts empty until you press Generate.

AREA	WHAT IT DOES
Sidebar (left)	Navigation: Roster, Coverage, Conflicts (red badge = number of hard issues), Changes, Setup, How it works. On a phone it becomes a scrolling bar at the top.
Page title	Today's date and the name of the page you're on.
Generate / Regenerate (yellow, top right)	Builds the month from the rules. Once a roster exists it becomes *Regenerate*, which rebuilds it but keeps locked manual edits.
Month selector ‹ December 2026 ›	Switches month. Each month keeps its own roster, version and change log.
Status pills	Draft · v1 roster state and version (goes up with every change) · 3 hard · 39 soft · 1 info live issue counts.
Repair	Adapts the existing roster to changes (approved leave, new rules) without rebuilding it. See chapter 15.
Publish (black)	Marks the roster as shared. Turns into *Edit (back to draft)* afterwards.
CSV	Downloads the roster and a coverage table for Excel / Google Sheets.

WHERE DATA LIVES

Everything is stored in **MongoDB** through the Embo server and saved automatically. Next to the Generate button you'll see **All changes saved**, **Saving…** or **Not saved — retrying**. Everyone using the same Embo address sees the same data. Use **Setup** → **Rules & data** → **Export JSON** for a backup.

CHAPTER 4

The sample data

The app ships with a realistic, fictional dataset so it can be tested immediately: **32 employees**, **4 pods** with a lead each, **25 brands**, **57 brand-processes** and **4 process types** (CS, ACC, NDR, RET).

POD	LEAD (FLOATER & BACKUP)	BRANDS
Alpha	Priya Nair	Bloom Botanics, UrbanThread, Kettle & Co, PawPantry, Glowveda (3 people), Nomad Gear (late shift)
Beta	Vikram Joshi	Crumb Bakehouse, VoltRide, SoleStory, FitFuel (2 people), LeafLuxe, Tiny Tots Co, HomeHearth
Gamma	Sana Sheikh	PureSip (3 people), Craftory (2 people), AquaPure Home, Spice Route, Rootsy
Delta	Neha Kapoor	Lumen Skincare, Brew Theory, Petal Post, Trailblaze (late shift), Mintly, Ozone Wear, Hearth & Hive (3 people)

Staffing patterns it demonstrates

PATTERN	WHERE TO SEE IT
80% single-owner brands	20 of 25 brands have one owner doing every process, e.g. Aman Sethi runs all 4 Lumen processes.
Separate processes	FitFuel: Karan does CS + NDR, Pooja does ACC + RET.
Shared process	PureSip: Aditya, Riya and Manav share CS, which needs 2 people a day (1 on Sundays).
Weekday minimums	Glowveda and Hearth & Hive CS need 2 people on Mondays.
One person, many brands	Arjun (Kettle & Co + PawPantry), Rahul, Yash, Divya, Kavya.
Late shift	Nomad Gear and Trailblaze: their backups can't also cover day-shift work that day.

Built-in scenarios (in the demo month)

Leave is placed relative to the month the app first opens on (December 2026 here):

SITUATION	WHAT EMBO DOES
Dev (VoltRide, sole owner, 365 days) on leave 7–9 Dec	Vikram covers CS on all three days; NDR is covered by Vikram on the 7th and Farah on the 8th–9th.
Siddharth (Trailblaze, sole owner, late shift) on leave 14–16 Dec **and** Neha, his only late-shift backup, on leave 15 Dec	**Unresolvable** on 15 Dec — flagged as two hard coverage gaps with an explanation.

SITUATION	WHAT EMBO DOES
Meera (Glowveda CS) sick on Mon 21 Dec, when CS needs 2	Tanvi plus backup Kabir keep Glowveda CS at 2.
Aditya (PureSip) on leave 22–24 Dec	Riya and Manav keep CS at 2.
Rahul uses a carried comp-off on 11 Dec	Shown as CO; comp-off balance tracked.
Kavya has a **pending** leave request 28–29 Dec	Shown as an info item waiting for a decision.
Christmas, Fri 25 Dec	365-day brands keep running (those staff earn a comp-off); other brands close.

CHAPTER 5

How a roster is generated

When you press **Generate**, Embo builds the whole month in under a tenth of a second using a fixed, explainable sequence. The same input always produces the same roster.

1. **Fixed facts first.** Locked (manually edited) cells are kept exactly. Approved leave becomes `L`, approved comp-off becomes `CO`. Everyone else starts as `W`.
   Why: these are commitments already made to people. The scheduler plans around them, never over them.
2. **Public holidays.** Everyone gets the holiday `H`. Then, for every brand-process that still operates (365-day ones), Embo pulls back the *minimum* number of owners needed, choosing whoever has worked the fewest holidays so far. Those people show `W★` and earn a comp-off.
   Example: on Christmas, Ananya (Crumb Bakehouse, closed on holidays) gets `H`; Rohan works Bloom CS (365 days) and earns a comp-off; PureSip CS needs 2, so Aditya and Riya work and Manav gets the holiday.
3. **Fill gaps caused by leave.** For each day and brand-process below its minimum, Embo picks the best trained backup (see "How a backup is chosen" below).
   Example: Dev is on leave 7–9 Dec, so Vikram and Farah cover VoltRide.
4. **Weekly offs.** Week by week (Monday–Sunday), each person gets their weekly off (normally 1). For every possible day, Embo *simulates* giving the off, fills any gap with backups, and scores the day (table below). The cheapest day wins.
   Order: first people whose max-consecutive-days deadline falls this week, then the most constrained people (sole owners of 365-day brands), then everyone else.
5. **Long-streak pass.** Working streaks can cross week boundaries, so a second pass checks for more than 6 days in a row and moves an off into the streak if coverage allows.
6. **Closed days.** If none of a person's own brand-processes operate on a day (and they aren't covering anyone), the day becomes `–`. Pod leads (floaters) always stay available.
7. **Final tidy-up.** One last gap fill, then any cover that is no longer needed is removed.

The weekly-off scoring table

Lower score = better day for the off. The numbers are deliberately far apart so that **client coverage always beats rest rules, which always beat preferences.**

IF THIS PERSON TAKES THE OFF ON THAT DAY…	SCORE	MEANING
a brand-process is left short and **no backup can fill it**	**+1000** per gap	Never chosen if any other option exists.
they would have worked **more than 6 days in a row** before it	**+300**	Labour rule.
a backup has to cover one of their processes	**+40** per process	Fewer handovers is better.
someone has to be called in from a public holiday	**+30** per person	Avoid spoiling someone's holiday.
it isn't their preferred off day	**+25**	Preference.
a co-owner of the same process is already off that day	**+6** each	Spread offs out.

IF THIS PERSON TAKES THE OFF ON THAT DAY…	SCORE	MEANING
it's a weekend and they already had weekend offs	**+4** × count	Rotate weekends fairly.

If **every** day of the week scores 1000 or more, Embo **does not force the off in**. It leaves the week without an off and the validator reports `Weekly off missing` with options — that is exactly what happens to Neha (Pod Delta lead) in the last week of December.

How a backup is chosen

A candidate must pass every check: has a BACKUP assignment for the process, is working that day (or on a holiday — a "call-in"), wouldn't exceed 6 brand-processes that day, and has **no shift clash** (can't mix GENERAL and LATE). Among those who pass, the lowest score wins:

```
score = (50 if called in from a holiday) + 2 × covers already done this month + 3 × processes already handled that day
```

WHY A GREEDY, RULE-BASED SCHEDULER (NOT AI OR A BLACK-BOX OPTIMISER)?

Every decision can be explained in one sentence ("Sunday was cheapest because ACC is closed and Rohan prefers Sundays"). It's deterministic, fast, and easy to test. The trade-off: it isn't guaranteed to find the mathematically best roster — which is why the separate validator and the repair tools exist.

CHAPTER 6

Worked example: why Rohan gets Sundays off

Rohan Mehta is the sole owner of **Bloom Botanics**: CS (365 days), ACC (Mon–Sat, closed on holidays) and NDR (every day except holidays). He prefers Sundays off. Backups: Sneha and Priya. Let's score the week **Mon 7 – Sun 13 December**, assuming his backups are free:

DAY	BLOOM PROCESSES OPEN	BACKUPS NEEDED	PREFERENCE	WEEKEND REPEAT	TOTAL
Mon 7	CS, ACC, NDR	3 × 40 = 120	+25	—	145
Tue 8 – Fri 11	CS, ACC, NDR	120	+25	—	145
Sat 12	CS, ACC, NDR	120	+25	+4 (had Sun 6 off)	149
Sun 13	CS, NDR (ACC closed)	2 × 40 = 80	0 (preferred)	+4	**84 ← chosen**

Sunday wins: one process is closed, and it's his preferred day. In the generated roster Rohan indeed has WO on Sun 6, Sun 13 and Sun 20, and on Sun 13 **Sneha covers Bloom CS and NDR** — open the Bloom Botanics brand view to see her green dots.

What if the backups were busy?

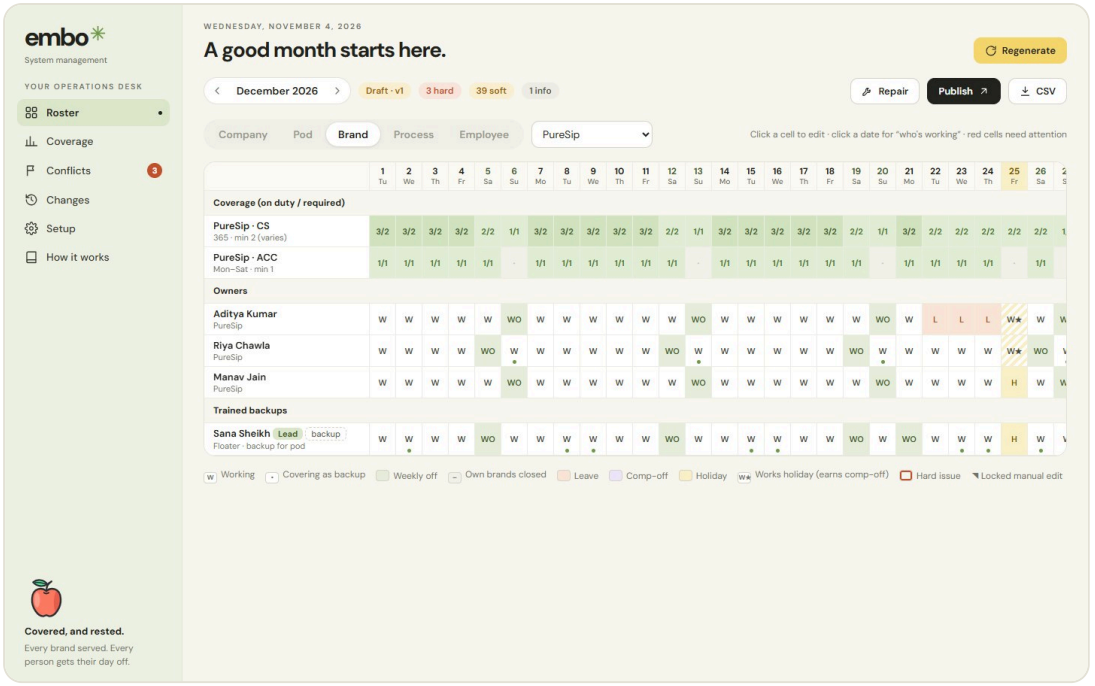
Suppose on Sunday both Sneha and Priya were already off or at their limit. Then Sunday's CS and NDR gaps can't be filled: score **2 × 1000 + …**. Monday–Saturday would also need backups; if one of those days has a free backup it scores ~145 and wins, so Rohan gets a weekday off — and the validator shows a soft Preferred off day missed. That is the "coverage beats preferences" rule in action.

REAL EXAMPLE FROM DECEMBER

Ananya (Crumb Bakehouse, CS open every day except holidays) prefers Sundays, but got Tue 1, Mon 7, Mon 14 and Tue 29 Dec. On those Sundays her backups (Nisha and Vikram) were already committed elsewhere, so a Sunday off would have left Crumb uncovered. The Conflicts page lists this as a soft issue: "preferred off day missed in 4 of 5 weeks … Client coverage took priority."

And the max-consecutive rule

If Rohan's previous off was Monday, a Sunday off the following week would mean working Tue → Sat of the next week (12 days in a row). The +300 penalty pushes the off earlier. If even that is impossible without an uncovered gap (+1000), the coverage wins and the validator raises Too many days in a row.



Brand view (PureSip). The coverage strip at the top shows "on duty / required" per day: 3/2 on weekdays, 1/1 on Sundays (weekday override), 2/2 during Aditya's leave.

CHAPTER 7

Validation: every rule it checks

The validator is a separate piece of code that never trusts the generator. It re-reads the roster cells and the configuration from scratch after **every** action: generation, a manual edit, a fix, a Setup change, a repair.

WHY A SEPARATE VALIDATOR?

So manual edits, imported data and generated rosters are judged by exactly the same rules — and so a bug or a limitation in the generator can never sneak an invalid roster past you. The automated tests confirm that the gaps the validator reports match the real gaps exactly.

HARD mandatory rules

RULE	TRIGGERED WHEN…	EXAMPLE / USUAL FIX
Coverage gap	A brand-process has fewer people on duty than required on an operating day.	"Trailblaze · CS short on Tue 15 Dec (0/1) — Sole owner: Siddharth Roy is on leave." Fix: assign a backup, move an off, add a backup.
No assigned employee	A brand-process operates but has no owner at all.	A new brand with no owner yet. Fix: Setup → Brands → add owner.
Leave conflict	Someone is rostered to work during approved leave.	You approved a leave in Setup. Fix: Repair.
Incompatible shifts	One person has GENERAL and LATE work on the same day.	Covering Trailblaze (LATE) while doing day-shift work.
Overloaded	More than 6 brand-processes in one day (own + covers).	Too many covers on one person.
Untrained cover	Covering a process they're neither backup nor cross-trained for.	Usually from imported/edited data.
Weekly off missing	A week (with 4+ days in the month) without the person's weekly off.	Neha, week of 28 Dec: she is the only backup for too many people.
Too many days in a row	A working streak longer than 6 days.	Move an off into the streak.
Comp-off balance	Carried + earned - used comp-offs is below zero.	Marking CO without having earned one.

SOFT preferences & fairness

RULE	TRIGGERED WHEN…
Preference not met	Weekly off fell on a non-preferred day. Reported once per person: “missed in 4 of 5 weeks”.
No weekend off	No Saturday or Sunday off all month although their brands run at weekends (e.g. Meera, Tanvi).
Cross-trained cover	Covered by someone cross-trained but not a designated backup.
Backup overused	Someone covers for others on more than 8 days (e.g. Neha: 17 days — a sign she needs help).

INFO needs a decision

RULE	TRIGGERED WHEN…
Pending leave	A leave request awaits approval for a rostered working day (Kavya, 28 Dec).
Multi-brand impact	One person’s absence leaves 2+ brands short on the same day.

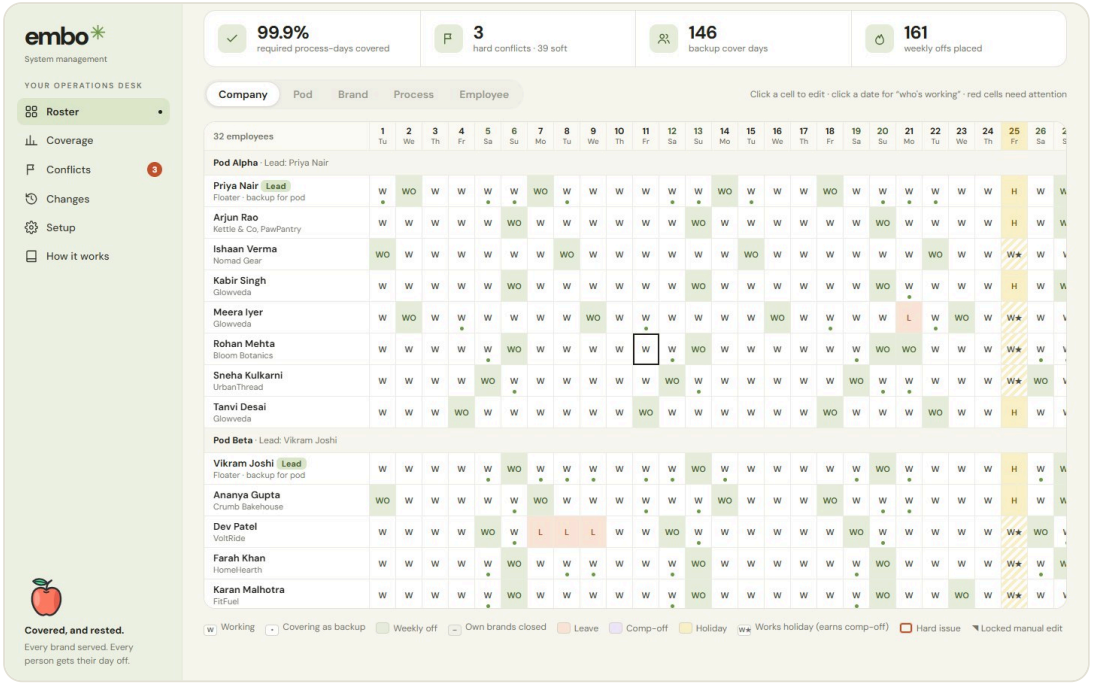
In December 2026 the counts are 3 hard (2 Trailblaze gaps + Neha’s weekly off), 39 soft (26 preference, 8 weekend, 5 backup-load) and 1 info (Kavya’s pending leave).

The Roster page

The main working screen. In **Company** view it starts with an overview, then the full grid.

The overview banner and tiles

ELEMENT	HOW IT'S CALCULATED	DECEMBER VALUE
Headline	"Every brand covered. Everyone rested." when there are no hard issues; otherwise "Almost there. N things need you." The link jumps to Conflicts (or Coverage).	3 things need you
Required process-days covered	(brand-process-days that meet their minimum) ÷ (all brand-process-days that need people) × 100.	99.9%
Hard conflicts · soft	Live count from the validator.	3 · 39 soft
Backup cover days	Person-days where someone covers at least one process for an absent owner.	146
Weekly offs placed	Number of `WO` cells (not counting "brands closed" days).	161

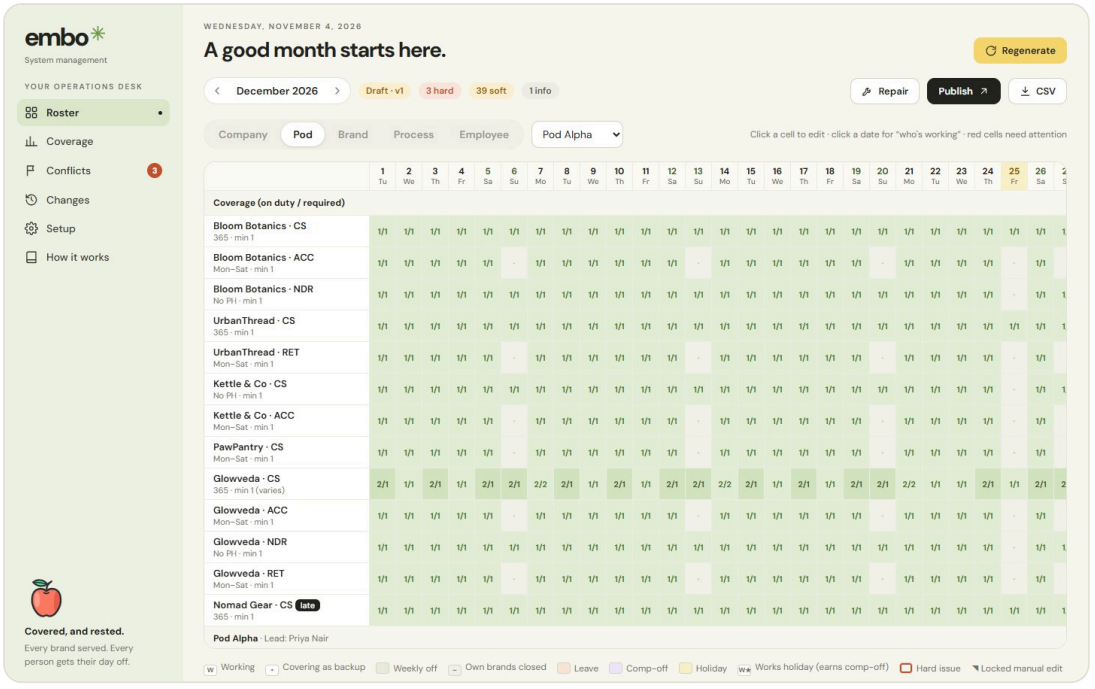


Company view grid. People grouped by pod, lead first. Day headers: weekends in green, the holiday in yellow. Each row shows the brands the person owns.

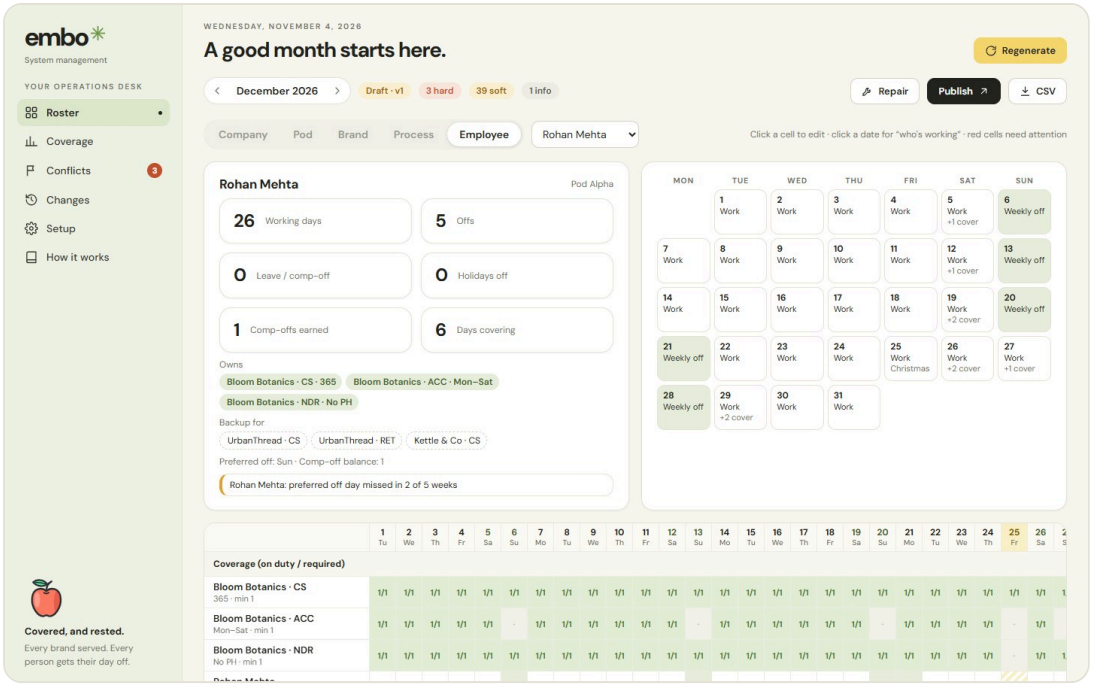
The five views

Use the pill switch above the grid. Views other than Company add a dropdown to pick the pod / brand / process / person.

VIEW	SHOWS	USE IT TO…
Company	Everyone, grouped by pod, plus the overview.	See the whole month at a glance.
Pod	The pod's people under their lead + a coverage strip for every brand-process in the pod.	A pod lead's daily check.
Brand	Coverage strip for the brand's processes, then owners and trained backups.	Answer a client: "are we covered?"
Process	Every brand's strip for one process (e.g. all CS) and the people who own it.	Compare staffing across brands.
Employee	One person's stats, responsibilities, calendar and issues.	Answer "when am I working?"



Pod view. Coverage strip per brand-process: green = met, dark green = above minimum, red = short (click for fixes), grey dot = closed.



Employee view (Rohan Mehta). Six tiles: working days, offs, leave/comp-off, holidays off, comp-offs earned (holidays worked), days covering. Below: what he owns, who he backs up, preferred off day, comp-off balance, and a Monday-first calendar — click any day to edit it.

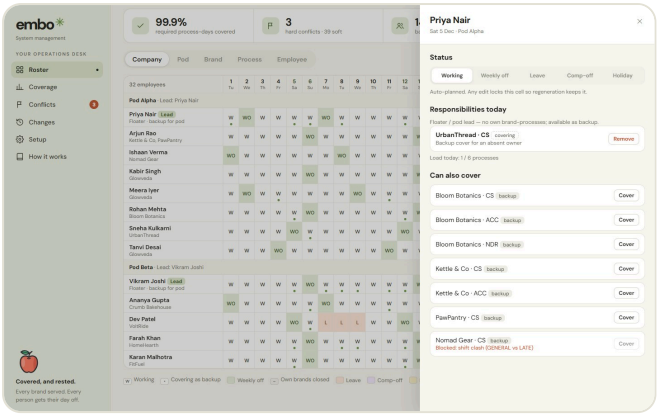
TRY IT

1. Roster → **Brand** → choose **Bloom Botanics**.
2. Find Sun 13 Dec: Rohan shows `WO`, Sneha shows a green dot — she's covering.
3. Click any person's name to jump to their Employee view.

CHAPTER 9

Editing a cell, and the day & gap panels

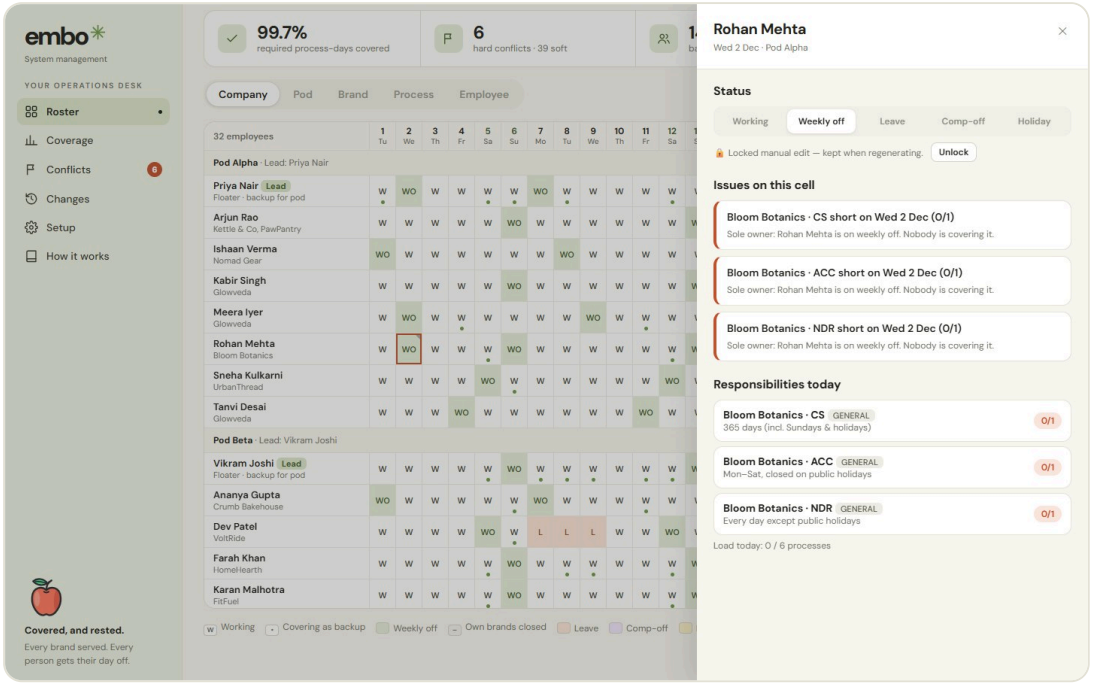
The cell editor (click any cell)



Cell editor for Priya Nair.

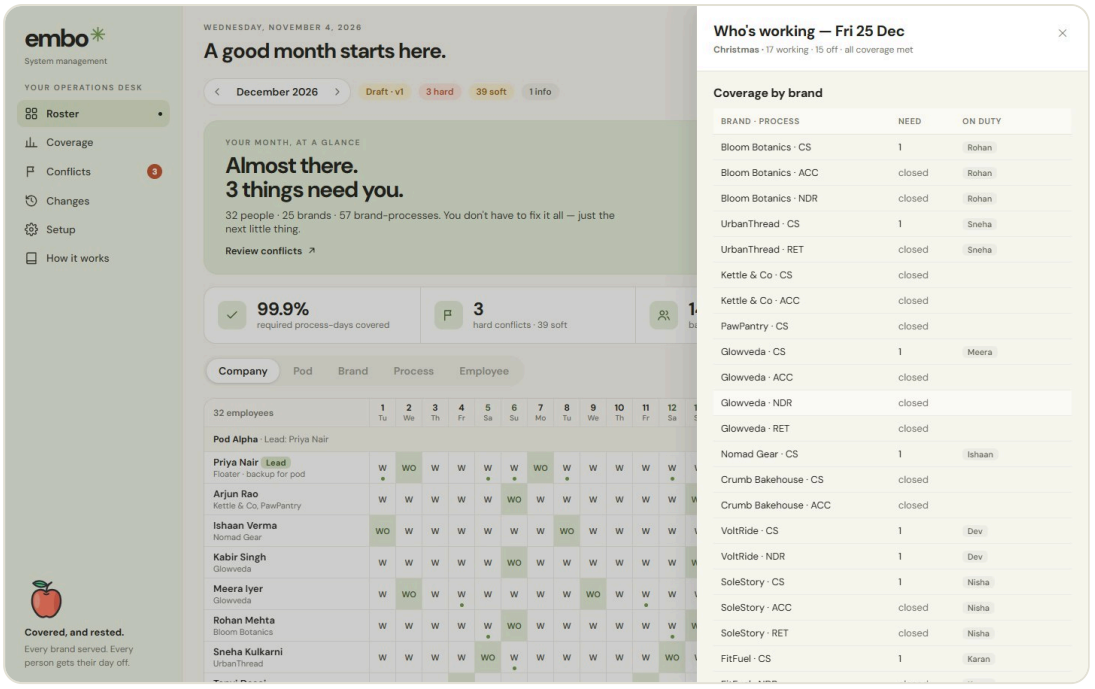
- **Status** — Working / Weekly off / Leave / Comp-off / Holiday. Changing it **locks** the cell (grey triangle) so Regenerate keeps it. *Unlock* hands the cell back to the scheduler.
- **Leave banner** — if the person has approved or pending leave that day.
- **Issues on this cell** — anything the validator found for this person/day.
- **Responsibilities today** — each owned process with its live coverage (e.g. 1/1), plus any covers, each with *Remove*. and a line showing today's load against the limit of 6.
- **Can also cover** — processes this person is trained for (backup or cross-trained), with short ones first. Blocked options say why ("shift clash", "would exceed 6").

Every edit shows a toast: *"1 new hard issue (4 total)"* or *"resolved 1 hard issue"*, so you always know the effect.



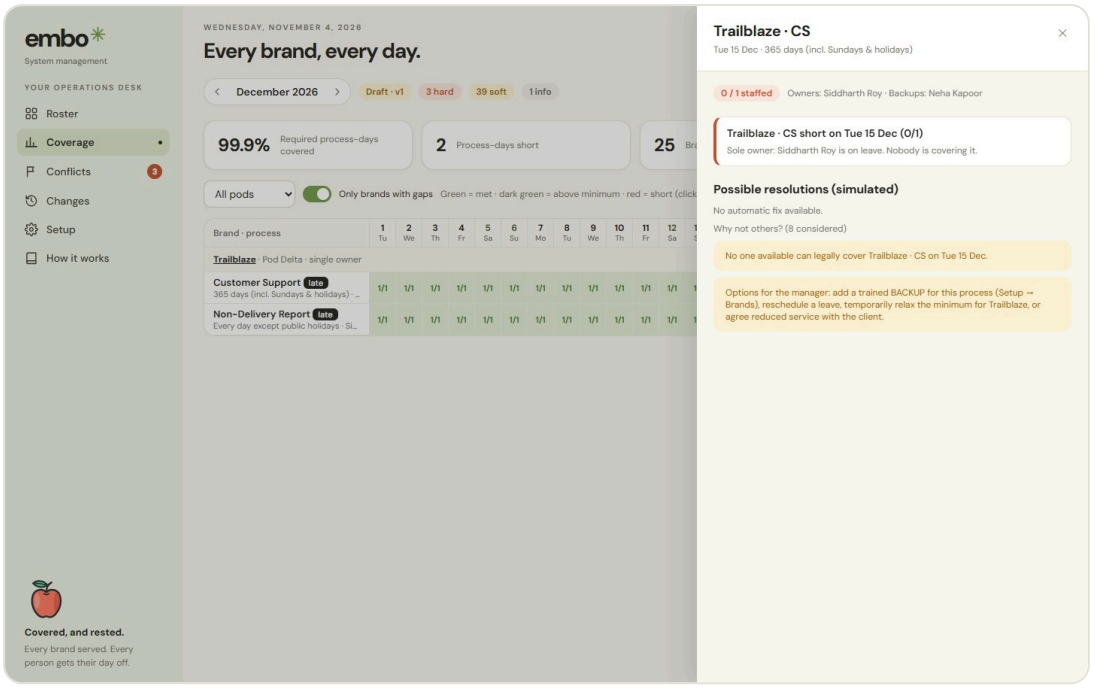
A manual edit creating a problem. Rohan is given a weekly off on a weekday: three Bloom processes are immediately flagged short, with the reason.

The “who’s working” panel (click a date header)



Christmas Day. Every brand-process with its requirement (closed ones say so) and who is on duty; dashed tags are backups. Short rows are clickable. Below: everyone not working and why — click a name to open their cell.

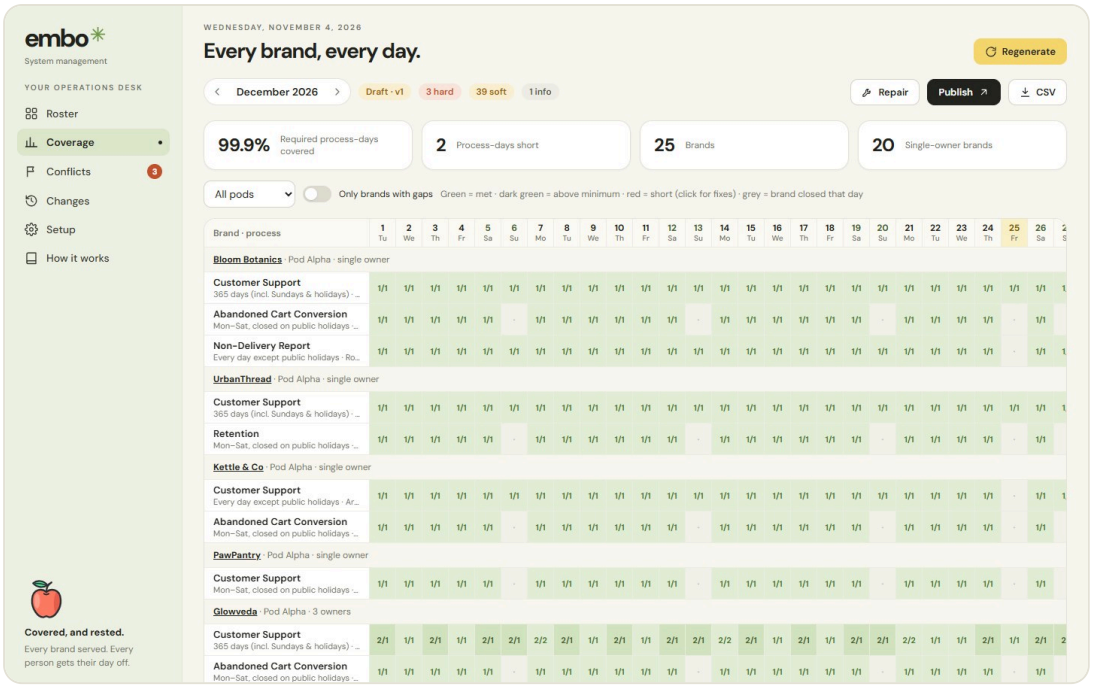
The gap panel (click a red coverage cell)



Gap panel. Staffed count, owners, backups, the issue and simulated resolutions (same engine as the Conflicts page).

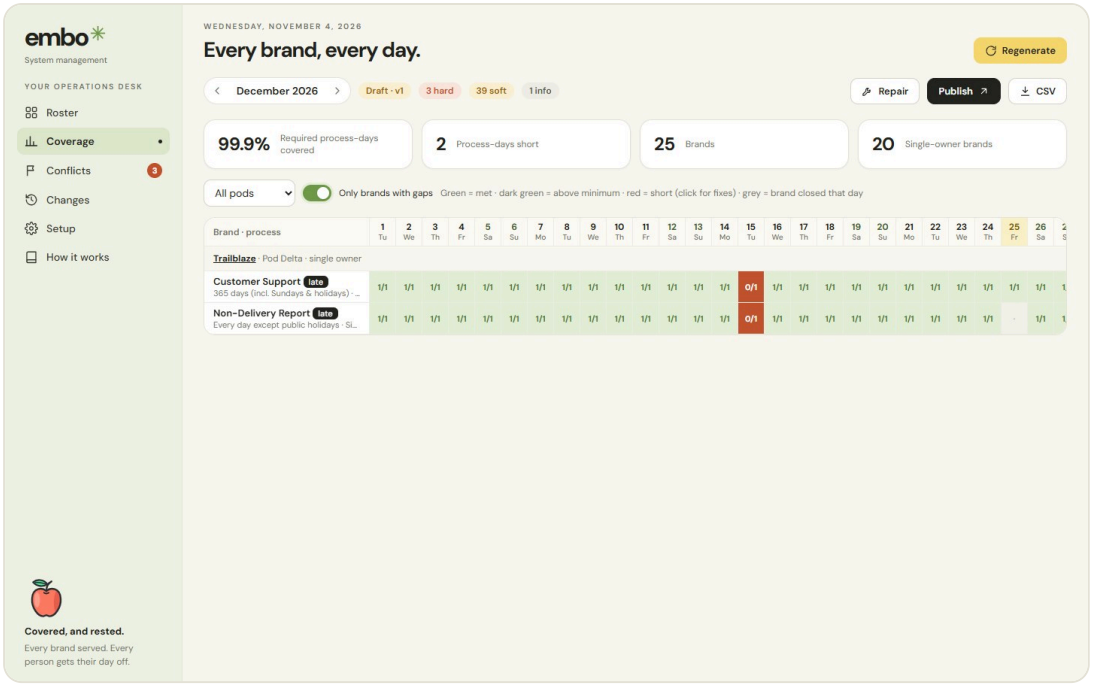
The Coverage page

The client's point of view: every brand, every process, every day — is the minimum met?



Coverage matrix. Each brand is a group; each row is one of its processes with its calendar and owners. Cells show "on duty / required".

ELEMENT	MEANING
Required process-days covered	Same formula as the Roster tile, but for the selected pod.
Process-days short	Count of red cells.
Brands / single-owner brands	How many brands are in view and how many depend on one person.
Pod filter	Limit to one pod.
Only brands with gaps (toggle)	Hide everything that's fine.
Cell colours	1/1 met · darker green = above minimum · 0/1 short · grey dot = closed that day.
Clicks	Red cell → gap panel with fixes. Any other cell or date → "who's working" panel. Brand name → Brand view.

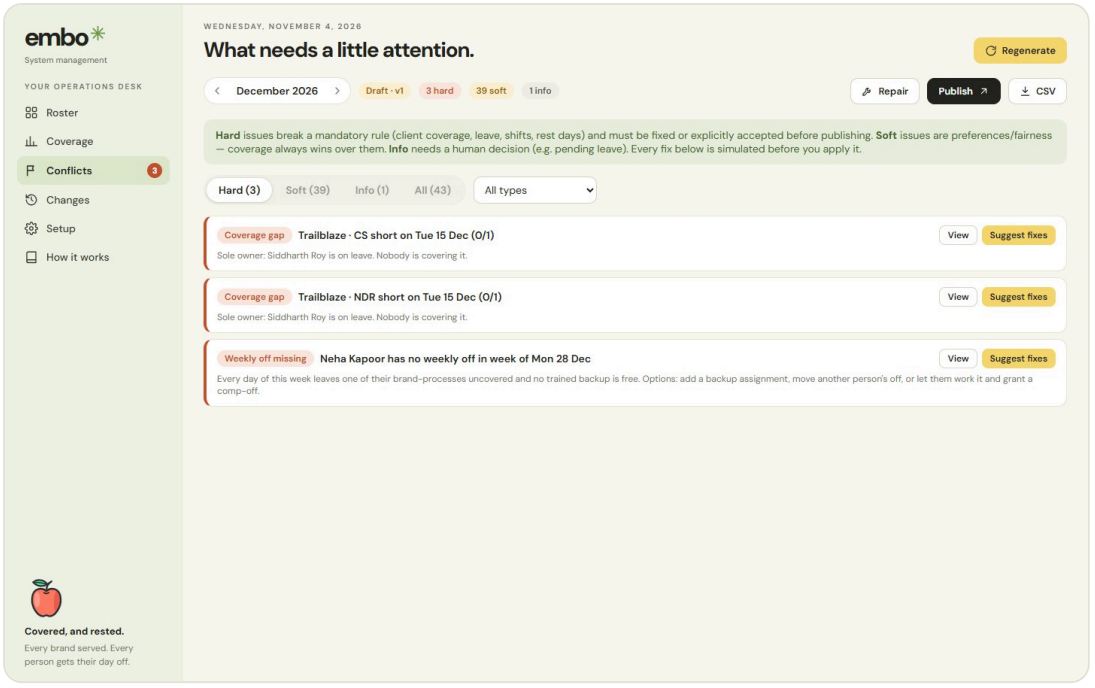


“Only brands with gaps” on. Just Trailblaze remains: red on the 15th for both processes.

READING DIFFERENT CALENDARS

Scroll to Lumen Skincare: CS is green every day (365), ACC is grey on Sundays and on Christmas (Mon–Sat, closed on holidays), NDR is grey only on Christmas (every day except holidays), RET follows ACC. One brand, three calendars.

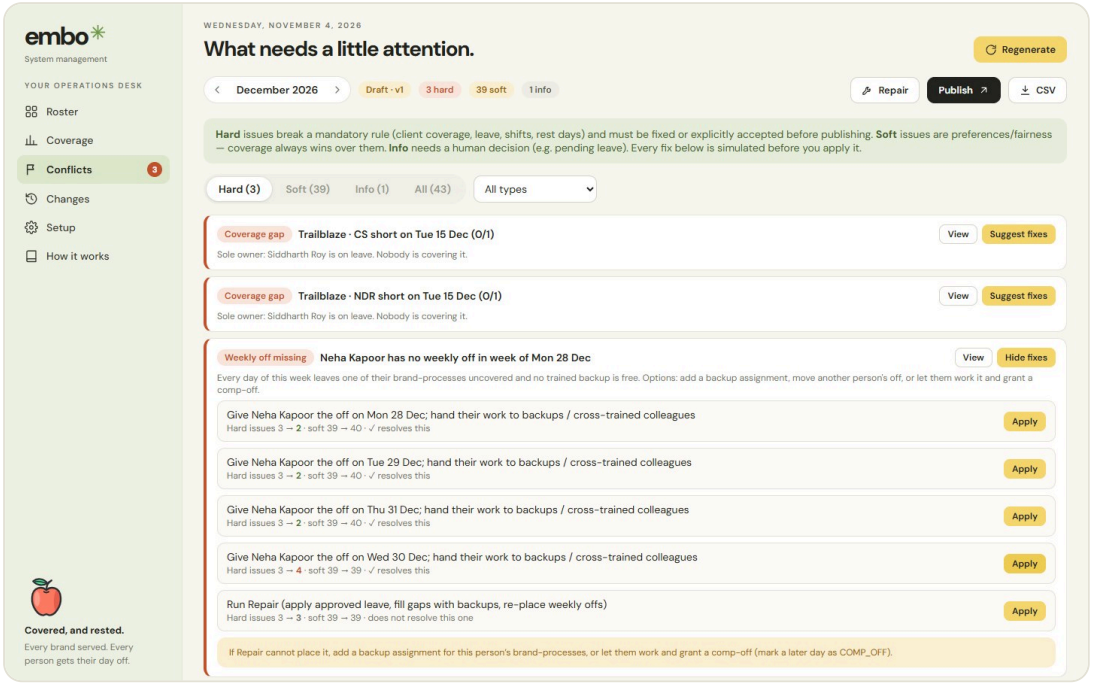
The Conflicts page & fixes



Conflicts, filtered to Hard. Tabs switch between Hard / Soft / Info / All; the dropdown filters by issue type. "View" jumps to the cell or brand.

"Suggest fixes" — simulated before you apply

For each issue Embo builds a list of possible fixes, **applies each one to a copy of the roster, re-validates**, and shows the result. You only commit when you press **Apply**.

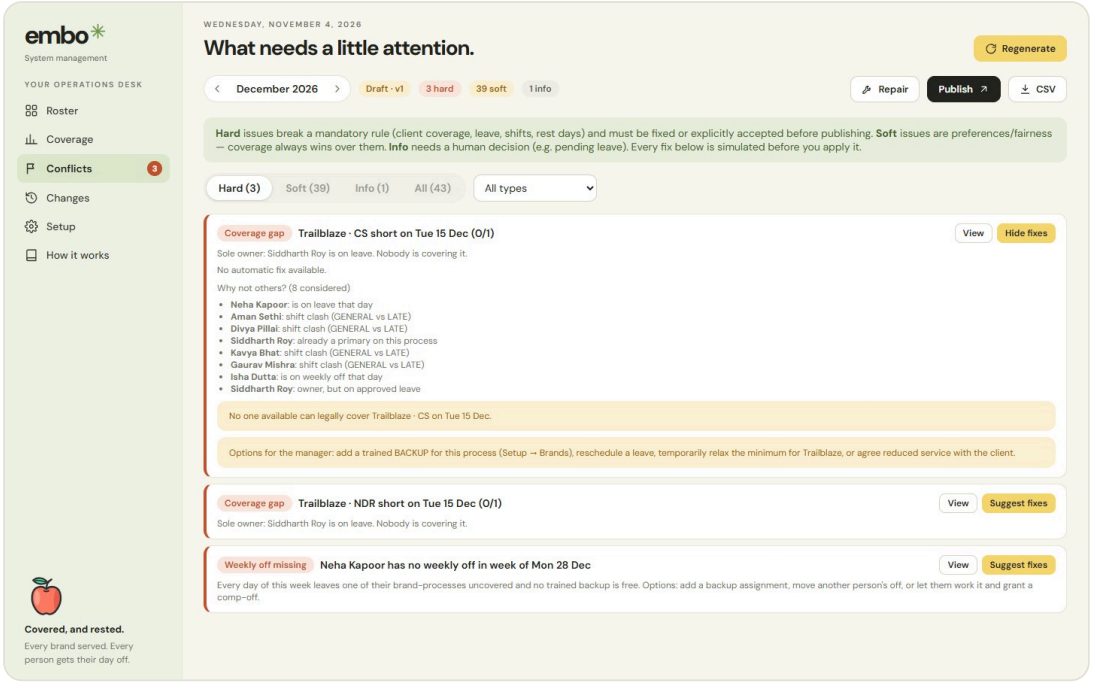


Fixes for Neha's missing weekly off. "Hard issues 3 → 2 · soft 39 → 40 · ✓ resolves this" means: applying it removes this problem and one hard issue overall, at the cost of one extra soft warning (a cross-trained colleague covers). The Wed 30 option would make things worse (3 → 4).

FIX TYPE	WHAT IT DOES	OFFERED FOR
Backup covers	A trained backup takes the process that day.	Coverage gaps
Cross-trained covers	A colleague from the same brand, or same process in the pod. Adds a soft warning.	Coverage gaps
Call in from holiday	Someone on a holiday works and earns a comp-off.	Coverage gaps on holidays
Move weekly off	An owner who is off that day moves the off to another day of the same week.	Coverage gaps
Give the off on day X	Places the missing weekly off and hands the person's work to backups/cross-trained colleagues.	Weekly off missing
Remove a cover	Takes a problem cover away (then Repair finds someone else).	Shift clash, overload, untrained
Run Repair	The general automatic adapter (chapter 15).	Leave conflicts, pending leave, missing offs

Applied fixes are manager decisions, so the cells they touch are **locked**.

When nothing can fix it



Trailblaze on 15 Dec. "No automatic fix available." Open "Why not others?" to see each person considered and the reason (on leave, shift clash, not trained). The yellow box lists the manager's real options: add a trained backup, reschedule a leave, relax the minimum, or agree reduced service with the client.

WHY NOT JUST LET IT AUTO-FIX EVERYTHING?

Some fixes have human costs (calling someone off a holiday, overloading a lead) or client costs (reduced service). Embo makes the trade-off visible and lets the manager decide.

The Changes page

Three tools for living with a roster after it is made, plus its history.

1 · Simulate a change (someone becomes unavailable)

Pick an employee, a date range and a type (sick, planned, comp-off), then **Preview impact**. Embo compares three worlds:

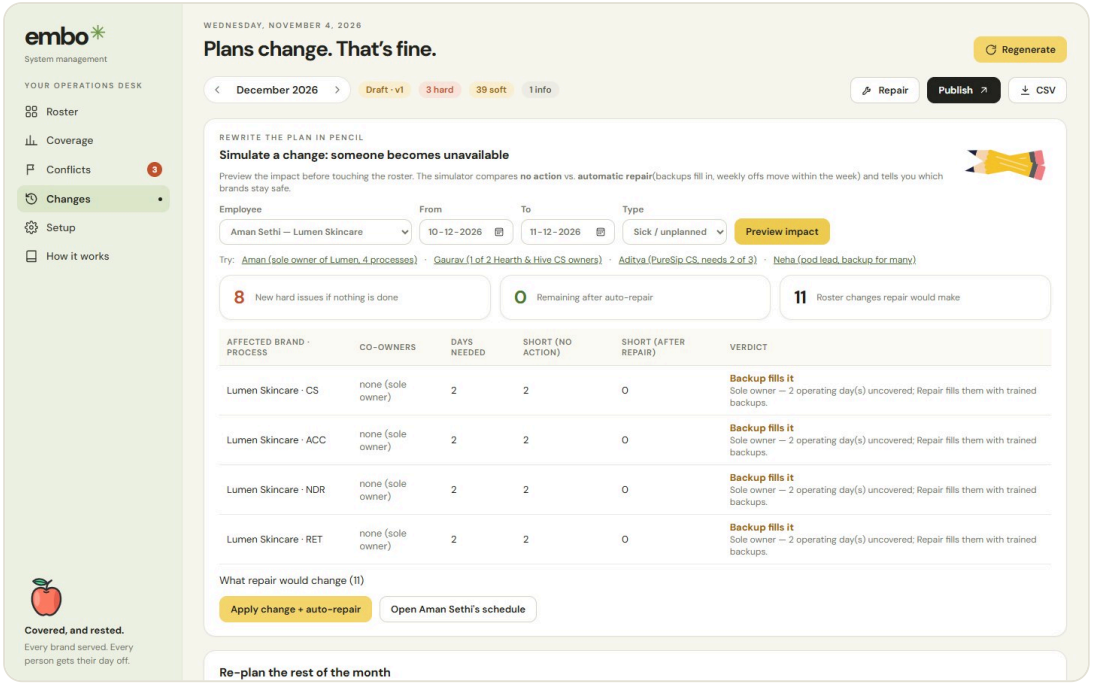
Current roster
As it is now.

→

No action
The person is marked absent and nobody reacts. Counts the new hard issues.

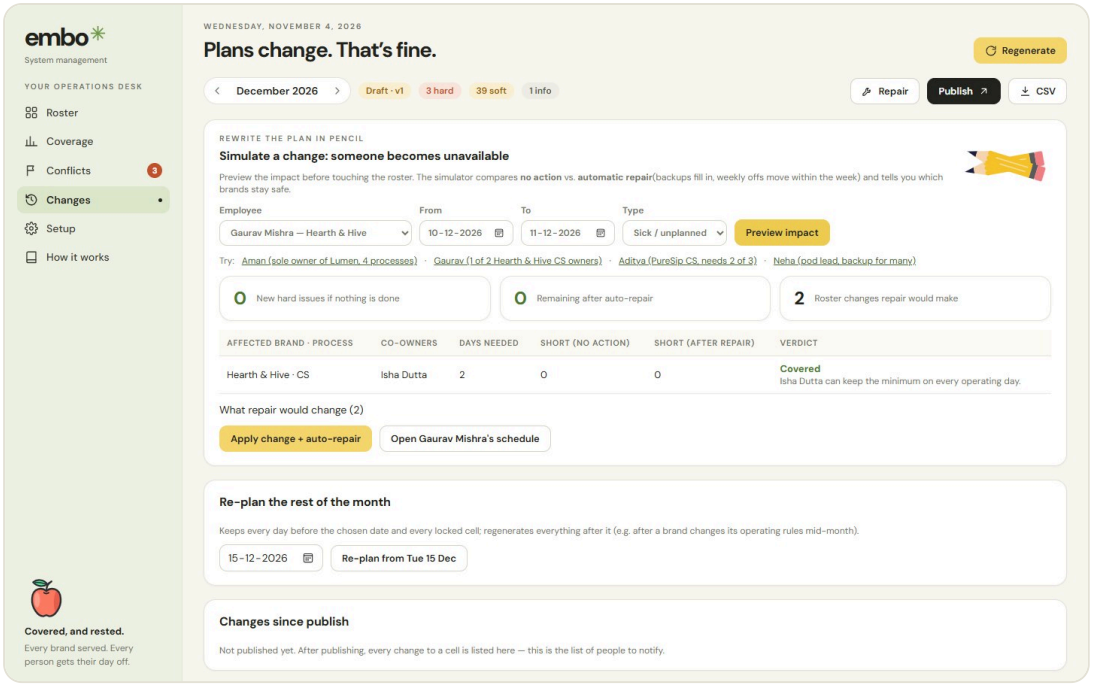
→

After auto-repair
Backups fill in, offs move within the week. Counts what's left.



Aman (sole owner of all 4 Lumen processes) sick on 10–11 Dec. 8 new hard issues if nothing is done; 0 after repair. Every Lumen process says "Backup fills it".

COLUMN	MEANING
Co-owners	Other owners of the process, or "none (sole owner)".
Days needed	Operating days in the window.
Short (no action / after repair)	Days below minimum in each world.
Verdict	Covered remaining owners can handle it · Backup fills it repair solves it · At risk needs a manager decision.



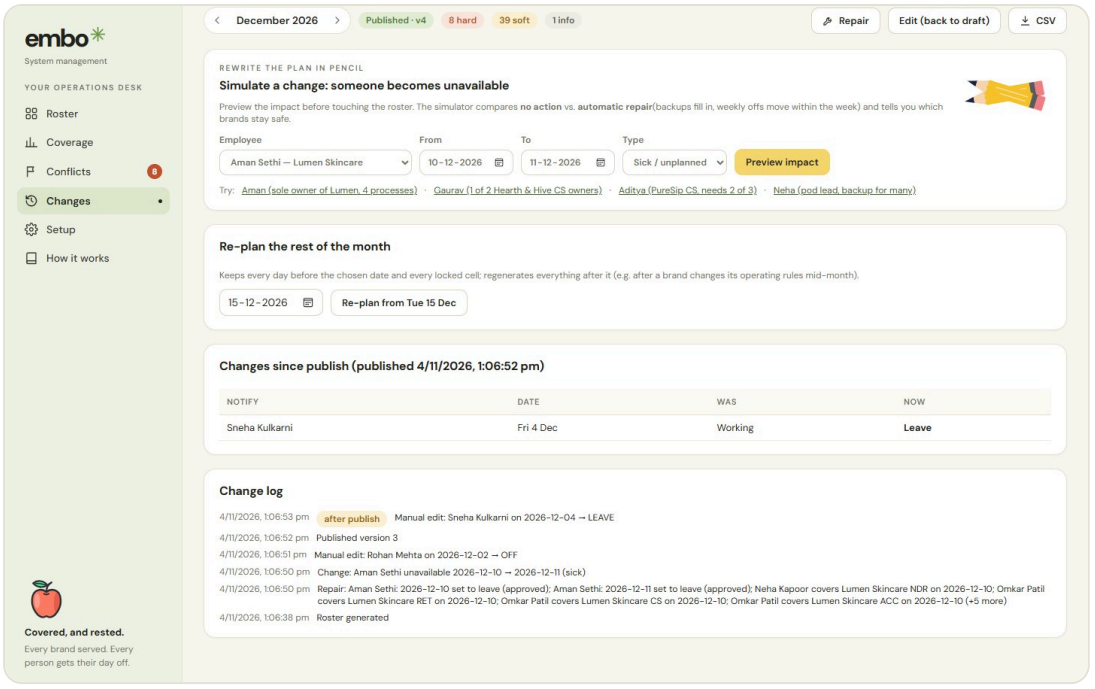
Gaurav (one of two Hearth & Hive CS owners) off Thu 10 – Fri 11 Dec. The remaining owner is evaluated: "Isha Dutta can keep the minimum on every operating day" → Covered. If the window included a Monday (when CS needs 2), that day would show "Backup fills it".

Apply change + auto-repair then adds the leave (as approved) to Setup, unlocks that person's cells in the window, runs Repair and writes it to the change log. The "Try" links pre-fill interesting people.

2 · Re-plan the rest of the month

Choose a date and press **Re-plan from …**. Everything *before* the date and every locked cell is kept; everything from the date onwards is regenerated. Use it when a brand changes its operating rules mid-month.

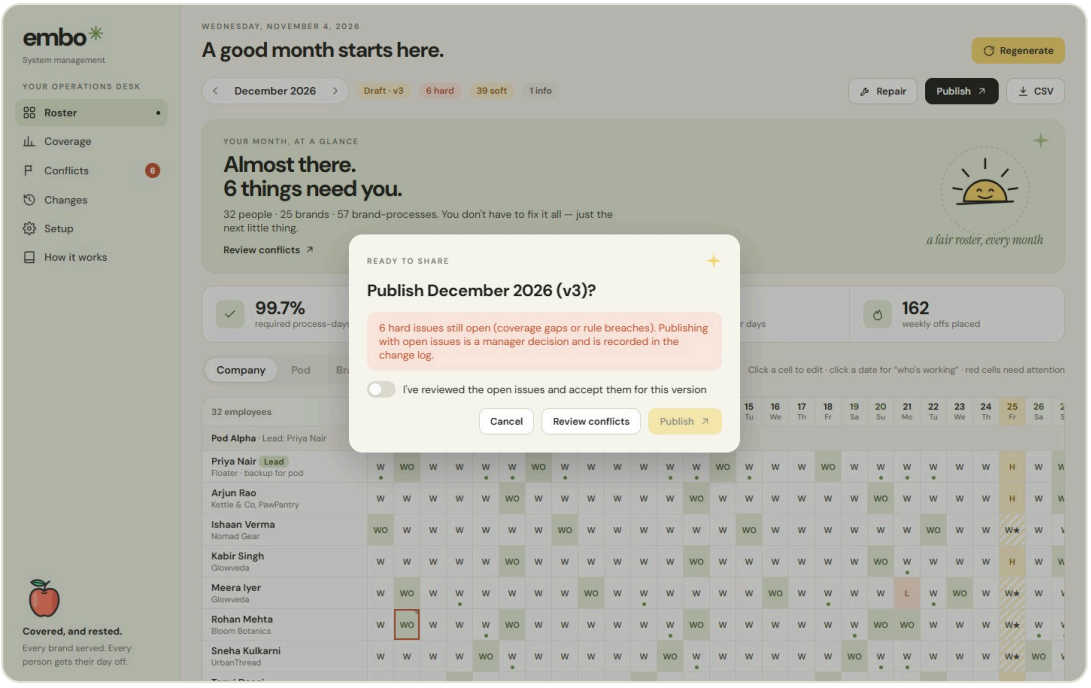
3 · Changes since publish & change log



After publishing, every cell that differs from the published version is listed (Notify · Date · Was · Now) — this is your notification list. Below, the change log records every action; entries after publishing are tagged.

CHAPTER 13

Publishing, notifications & export



Publishing with open hard issues requires ticking "I've reviewed the open issues and accept them". With no hard issues it just confirms.

- **Publish** saves a snapshot of the roster. The pill changes to `Published · vN`.
- You can still edit a published roster. Each changed cell gets an orange dot in the grid and appears under **Changes since publish**. **Edit (back to draft)** marks it as a draft again.
- The **version** number increases with every change, so "v7" tells everyone which edition they're looking at.

CSV export

The **CSV** button downloads `roster-YYYY-MM-vN.csv` with:

- One row per person: name, pod, owned processes, one column per day with the code (covers listed in brackets), then totals: working days, weekly offs, leave, comp-offs earned.
- A coverage table below: every brand-process with "on duty/required" per day, or "closed".

Open it in Excel or import it into Google Sheets (File → Import).

> **WATCH OUT**
>
> Publishing does not send emails or messages; the "Changes since publish" list tells you who to contact.

CHAPTER 14

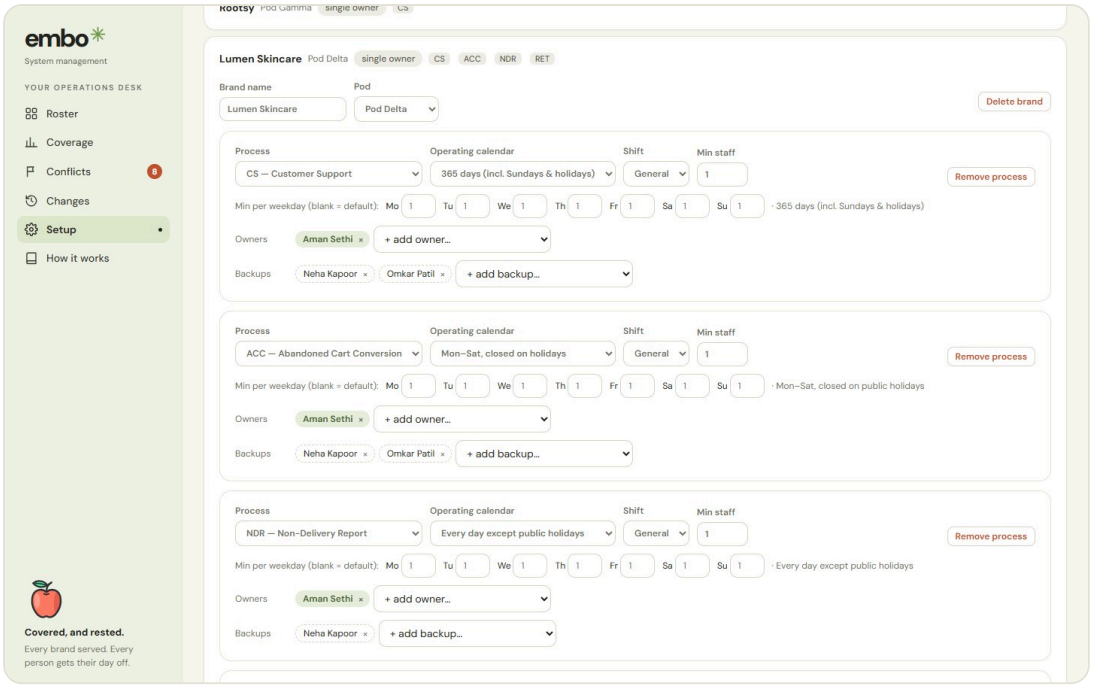
Setup: changing everything

Everything the scheduler uses is data you can change here — no code. Changes are validated against the current roster **immediately** (watch the hard/soft pills), but the roster itself is **not rewritten** until you press Repair or Regenerate.

WHY DOESN'T A SETUP CHANGE REWRITE THE ROSTER?

So nothing changes behind your back. You see the consequence (a new conflict) first, then decide how to adapt: Repair (small, local) or Regenerate (rebuild).

Brands (processes & assignments)



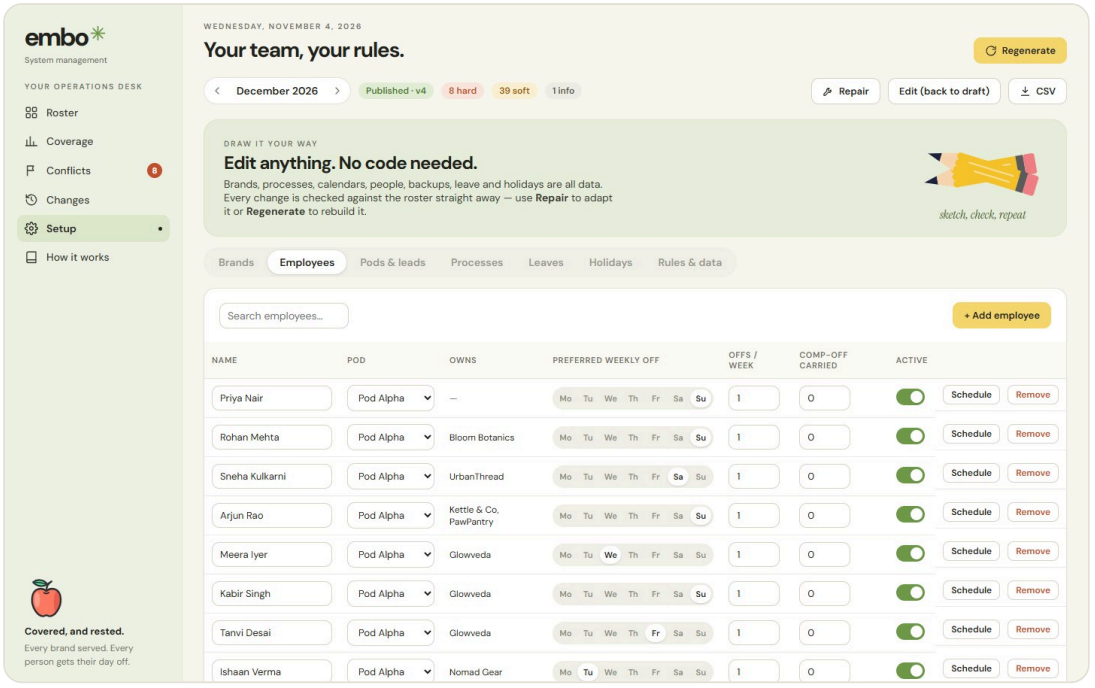
An expanded brand card (Lumen Skincare). Each process has its own calendar, shift, minimum staff, per-weekday minimums, owners and backups.

CONTROL	WHAT IT DOES
Search / pod filter	Find brands quickly.
+ Add brand	Name + pod. Starts with Customer Support (365 days, min 1).
Brand name, Pod, Delete brand	Rename, move between pods, or delete (removes its processes and assignments).
Process	Which process type this row is (CS, ACC…).
Operating calendar	365 / every day except holidays / Mon–Sat / Custom. Custom shows a weekday picker and a "closed on holidays" switch.
Shift	General or Late.

CONTROL	WHAT IT DOES
Min staff	Default people needed.
Min per weekday	Leave blank to use the default; type a number to override that weekday (e.g. Mo = 2).
Owners / Backups	Chips with × to remove; "+ add owner / backup" dropdown lists active people (same pod first).
+ Add process to …	New process row (default Mon–Sat, min 1).

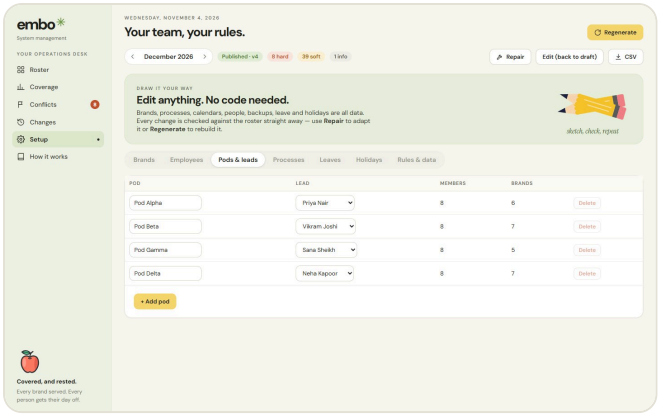
EXAMPLE: MAKE BLOOM CS NEED 2 PEOPLE ON SATURDAYS

1. Setup → Brands → expand Bloom Botanics.
2. On the CS row, type **2** in the **Sa** box.
3. The header jumps to more hard issues: Bloom CS is now short every Saturday (Rohan is the only owner).
4. Add a second owner or more backups, then press **Repair** — backups are assigned to the Saturdays.

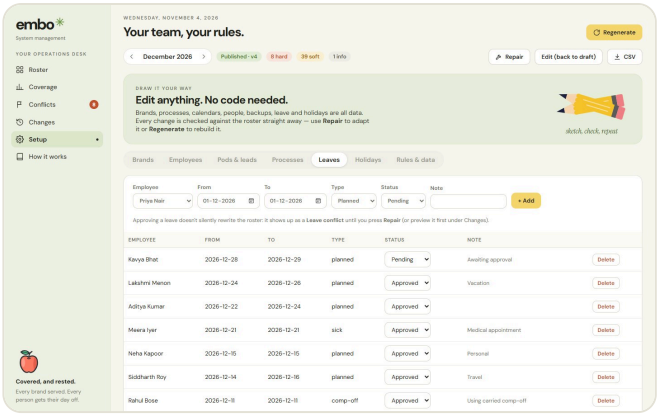


Employees. Name, pod, what they own (read-only), preferred weekly off (pill weekday picker), offs per week, comp-offs carried in, active switch, Schedule (opens their view), Remove.

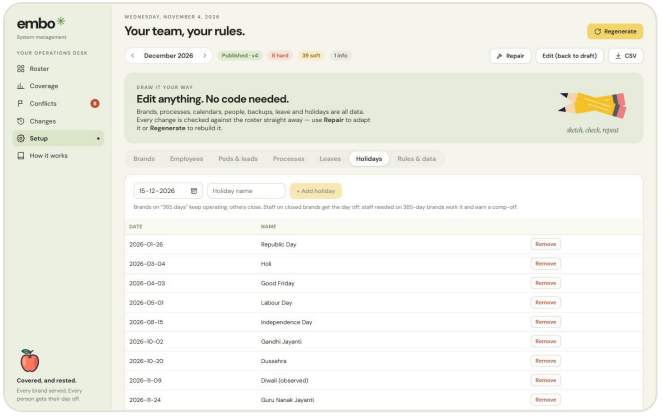
- **Active switch off** = the person disappears from planning (e.g. long leave or left the company) but their data stays.
- **Remove** deletes the person, their assignments and leave, and clears them as pod lead.
- **Offs / week** can be 0–3 (e.g. 2 for a part-timer).



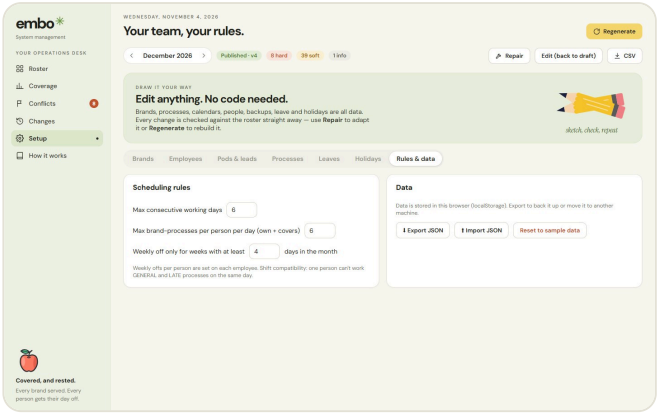
Pods & leads. Rename, pick the lead from the pod's members, add pods. Delete only works on an empty pod.



Leaves. Add (employee, dates, type, pending/approved, note). Change status in the list: Pending → Approved / Rejected.



Holidays. Add a date + name; this month's holidays are bold.



Rules & data. Max consecutive days (6), max processes per person per day (6), minimum days in a week for a weekly off (4). Export / Import JSON, Reset to sample data (affects everyone using the database).

Process types tab: rename codes and names, add new process types (e.g. "Social Media"), delete unused ones.

CHAPTER 15

Generate vs Regenerate vs Repair vs Re-plan

ACTION	WHAT IT CHANGES	WHAT IT KEEPS	USE WHEN…
Generate	Creates the month from scratch.	—	The month has no roster yet.
Regenerate	Rebuilds every unlocked cell.	Locked manual edits; version history and log.	Big rule changes early in planning.
Repair	Only what is broken (see below).	Everything that is still valid, unchanged.	After approving leave, editing a rule, or any small change. **Your default.**
Re-plan from date	Every unlocked cell from that date on.	All earlier days + locked cells.	Mid-month changes to brand rules.

What Repair actually does, in order

1. **Syncs leave:** approved leave → `L` / `CO`; cancelled or rejected leave → back to `W`.
2. **Cleans up:** removes covers on non-working cells or for processes that are now closed; turns "brands closed" days back to work if a brand now operates.
3. **Re-checks holidays** (only pulling people back if needed).
4. **Fills gaps** with trained backups.
5. **Moves weekly offs within the same week** to close remaining gaps — including a *chain move*: if the only backup is busy covering someone else's day off, it moves *that* person's off.
6. **Places missing weekly offs**, re-marks closed days, fills gaps once more and removes unneeded covers.

Repair never touches locked cells, and every action it takes is listed in the change log.

CHAPTER 16

Recipes: common situations, step by step

Someone calls in sick for tomorrow

1. **Changes** → pick the person and the dates → type “Sick” → **Preview impact**.
2. Read the verdicts. All “Covered / Backup fills it” → **Apply change + auto-repair**.
3. Any “At risk” → apply anyway, then open **Conflicts** and choose a fix or a manager option.

Approve a leave request

1. Setup → **Leaves** → set status to **Approved**.
2. A Leave conflict appears (the roster still has them working).
3. Press **Repair**. Tip: preview it first under Changes.

A client now wants Sunday support

1. Setup → Brands → the brand → change that process’s calendar from Mon–Sat to **Every day except public holidays** (or 365).
2. New gaps appear for Sundays (or the “closed” days come back).
3. Press **Repair** (or **Re-plan from** the date the change starts).

A new hire joins and takes over a process

1. Setup → Employees → **+ Add employee**; set name, pod, preferences.
2. Setup → Brands → the brand → add them as **owner** (and as backup elsewhere).
3. **Repair** — they get weekly offs and their coverage counts.

Someone wants Saturdays off from now on

1. Setup → Employees → set their preferred weekly off to **Sa**.
2. **Regenerate** (preferences are applied when offs are placed). If coverage allows, they get Saturdays; otherwise a soft issue explains why not.

A pod lead is overloaded (backup overused)

1. Look at Soft → “Backup overused” (e.g. Neha covers 17 days).
2. Setup → Brands → add a **second backup** to the processes she covers.
3. **Regenerate** — load spreads across backups.

Add a company holiday

1. Setup → Holidays → date + name → Add.
2. **Repair**: 365-day brands keep someone on (comp-off earned); others close.

Fix a single day by hand

1. Click the cell → change status or add a cover from “Can also cover”.
2. Read the toast; open Conflicts if new issues appeared.

Publish and tell people what changed

1. Publish (acknowledge any open issues).
2. After later edits: Changes → **Changes since publish** = who to notify. Export CSV to share.

Back up, move or reset your data

1. Setup → Rules & data → **Export JSON** (save the file).
2. To restore, or to move to another Embo server: **Import JSON**.
3. **Reset to sample data** starts over (for everyone).

CHAPTER 17

Under the hood

Embo is a **MERN** app written in TypeScript: **React 18** (Vite, Zustand) in `client/`, an **Express + Mongoose** API in `server/`, **MongoDB** for storage, and the scheduling engine in `shared/`, used by both. The server runs Generate, Re-plan and Repair and saves the results. The browser uses the same engine for instant checks and fix previews. Edits are saved automatically through REST calls. One Node service (e.g. on Render) serves both the API and the React app. See `DEPLOYMENT.md` for hosting.

FILE	RESPONSIBILITY
`shared/src/engine/types.ts`	The data model (pods, employees, brands, brand-processes, assignments, leave, roster cells, issues).
`shared/src/engine/calendar.ts`	Date maths, the four calendars, required staff per day.
`shared/src/engine/context.ts`	Fast lookups, `coverage()`, `canCover()`, responsibilities.
`shared/src/engine/generate.ts`	`generateRoster()` and `repairRoster()`.
`shared/src/engine/validate.ts`	Every hard/soft/info rule.
`shared/src/engine/suggest.ts`	Fix suggestions and their simulation.
`shared/src/engine/impact.ts`	The change simulator.
`shared/src/data/seed.ts`	Sample dataset.
`server/src/`	Express app, Mongoose models, REST routes, services, seed script.
`client/src/store/`	App state, loading from and autosaving to the API.
`client/src/ui/`	Pages, panels, styles, CSV/JSON export.

WHY THE ENGINE IS SEPARATE FROM THE SCREENS

The engine is plain functions with no screen or storage code. That makes it testable (17 automated tests) and portable: it could run on a server unchanged if Embo moves to a shared database.

Running it

COMMAND (IN THE PROJECT FOLDER)	DOES
`npm install`	Installs dependencies (first time).
`npm run dev`	Starts the API (port 4000) and the app (http://localhost:5173). Needs `server/.env` with `MONGODB_URI`.
`npm run seed`	Reloads the sample data into MongoDB.
`npm test`	Runs the engine tests.

COMMAND (IN THE PROJECT FOLDER)	DOES
`npm run build`	Builds the React app and the bundled server; `npm start` then runs both on one port.

The tests prove, for four different months, that every real gap is reported, that the only gap left is the designed Trailblaze case, and that the generator never breaks shift, overload, leave, untrained-cover, max-consecutive or comp-off rules. Others check holidays, PureSip's minimum of 2, locks surviving regeneration, manual-edit detection, sole-owner and shared-owner absences, repair leaving untouched days identical, and the unresolvable explanation.

CHAPTER 18

Assumptions & limits

Assumptions (configurable unless noted)

1. One shift per person per day; processes are GENERAL or LATE and can't be mixed in a day.
2. Weeks run Monday–Sunday. A week earns a weekly off only if at least 4 of its days fall in the month.
3. At most 6 consecutive working days (setting). Streaks across months are not tracked.
4. At most 6 brand-processes per person per day, own + covers (setting).
5. One company-wide holiday list. Working a holiday earns 1 comp-off; balance = carried + earned - used.
6. A public holiday is never used as a weekly off — they are separate entitlements.
7. Leave doesn't count as a weekly off; a week with enough leave needs none.
8. Only designated backups are used automatically; cross-trained colleagues are suggested only.
9. Pending leave doesn't change the roster; approved leave applies on Repair.
10. If all of a person's brands are closed on a day, that day is off for them (floaters stay available).

Known limits

- No logins yet: anyone with the address can edit. If two people edit at the same moment, the last save wins.
- Each month is planned independently.
- Two shift types only.
- The scheduler is fast and explainable but not guaranteed optimal — the validator always tells you exactly what is wrong.
- No automatic emails/messages on publish.

CHAPTER 19

FAQ: “why did it do that?”

Why did someone get a weekday off instead of their preferred Sunday?

Their Sunday off would have left a process uncovered (no free backup) or cost more backups. Coverage outranks preferences — see the scoring table in chapter 5. The soft issue lists the weeks it happened.

Why isn’t the Trailblaze gap fixed automatically?

Nobody can legally do it: the owner and the only late-shift backup are both on leave, and other colleagues work the day shift (shift clash). Embo won’t break a hard rule to hide a gap.

I approved a leave but the roster still shows them working.

By design. You’ll see a Leave conflict. Press Repair (or preview under Changes first).

Why does Neha have no weekly off in the last week?

She is the only trained backup for several people in that 4-day week; every day she’d take off leaves someone uncovered. Fix: apply a suggested “Give the off” option, or add another backup for her processes.

What does “–” mean?

All of that person’s own brands are closed that day and their weekly off is already elsewhere, so they are off anyway. In the December sample you won’t see many: for Mon–Sat brands like Rahul’s, the closed Sunday is simply used as the weekly off (`WO`), which costs nothing in the scoring.

Why did my manual change survive “Regenerate”?

Manual edits lock the cell. Use **Unlock** in the cell editor to let the scheduler decide again.

The numbers on my screen differ from this guide.

You’ve edited data, applied fixes, or you’re on a different month. Setup → Rules & data → Reset to sample data, then Generate December 2026.

Can two people see the same roster?

Yes. Everyone using the same Embo address reads and writes the same MongoDB database. Reload the page to see someone else’s latest changes.

CHAPTER 20

Glossary

Brand-process	One process for one brand (e.g. Lumen · CS); the unit that needs coverage.
Owner (primary)	Person assigned to do a brand-process; counts automatically when working.
Backup	Trained stand-in; counts only when given a cover.
Cross-trained	Same brand on another process, or same process in the same pod; suggested only.
Cover	A backup taking on a process for a day (green dot).
Floater	Someone with no own processes (the pod leads) — always available as backup.
Operating calendar	The days a brand-process runs.
Minimum staff	People required on an operating day (with weekday/date overrides).
Coverage	People on duty for a brand-process on a day, shown as on duty/required.
Weekly off (WO)	Rest day per Monday–Sunday week.
Comp-off (CO)	Day off earned by working a public holiday.
Hard / Soft / Info issue	Mandatory rule broken / preference or fairness / needs a decision.
Lock	A manually edited cell the scheduler won't change.
Repair	Local, non-destructive adaptation of the roster to changes.
Re-plan	Regenerate from a chosen date onwards.
Version	Counter that increases with every roster change.

one month at a time

Embo — System management